Two four vectors a, b

$$a \cdot b = a_0 b_0 - \vec{a} \cdot \vec{b} = a_0 b_0 - a_1 b_1 - a_2 b_2 - a_3 b_3$$

$$a \cdot b = a_\mu b^\mu = a_\mu g^{\mu\nu} b_\nu = g^{\mu\nu} a_\mu b_\nu$$

$$g^{\mu\nu} = \begin{pmatrix} +1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$$

$$\begin{array}{cccc} 0 & 1 & 2 & 3 \end{array}$$

$$e = (E_e, 0, 0, P_z), \quad e^2 = m_e^2 \simeq 0 = E_e^2 - 0^2 - 0^2 - P_z^2$$

thus $P_z = E_e$

so

$$e = (E_e, 0, 0, E_e)$$

$$e' = (E_{e'}, \qquad), \quad e'^2 = 0 = E_{e'} - \vec{E}'^2 \text{ thus}$$

$$|\vec{E}'| = E_e$$

$$\vec{r} = (r_1, r_2, r_3)$$

$$\vec{r} = r(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta) \quad \text{spherical } (r, \theta, \varphi)$$

$$\vec{r} \cdot \vec{r} = r^2, \quad \text{„}r\text{"} = E_{e'}$$

$l' = (E_{l'}, E_{l'} \sin\theta \cos\phi, E_{l'} \sin\theta \sin\phi, E_{l'} \cos\theta)$

<u>let $\phi = 0$</u> you have defined <u>your</u> "x"

$l' = (E_{l'}, E_{l'} \sin\theta, 0, E_{l'} \cos\theta)$

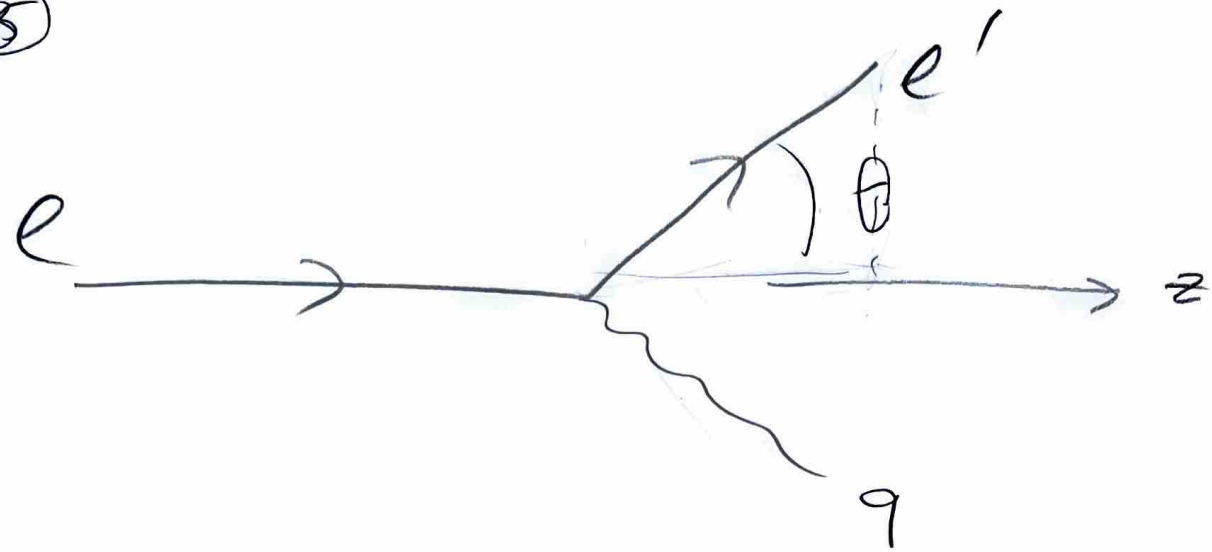
$l \cdot l' = E_l E_{l'} - 0 - 0 - E_{l'} \cos\theta \, E_l$

$q^2 = -2 E_l E_l' (1 - \cos\theta) \leq 0$

so

$q^2 = -Q^2$ where $Q^2 = 2 E_l E_{l'} (1 - \cos\theta) \geq 0$

(3)



$$\ell = (E_e, 0, 0, P_3) \qquad \ell^2 = m_e^2 \simeq 0$$

$$\ell'^2 = m_e^2 \simeq 0$$

$$\ell' = (E_{e'}, P_1', P_2', P_3')$$

$$\ell = \ell' + q \qquad q = \ell - \ell'$$

$$q^2 = (\ell - \ell')^2 \quad , \quad \underline{q^2 < 0} \, .$$

Real

why $q^2 < 0$:

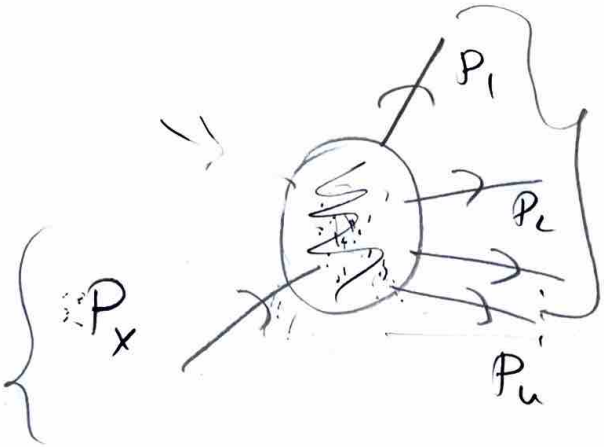
$$q^2 = (\ell - \ell')^2 = \underbrace{\cancel{\ell^2}}_{\simeq 0} - 2\,\ell\cdot\ell' + \underbrace{\cancel{\ell'^2}}_{\simeq 0} = -2\left[E_e E_{e'} - 0 - 0 - P_3 P_3'\right]$$

$$= -2\left(E_e E_{e'} - P_3 P_3'\right)$$

since leptons have mass, $v < c$
so $E_e E_{e'} > P_3 P_3'$ and $q^2 < 0$

$$c^2 (m_e \gamma)(m_e \gamma') > (m_e \gamma v_\ell)(m_e \gamma' v_{\ell'})$$

(including c for clarity)



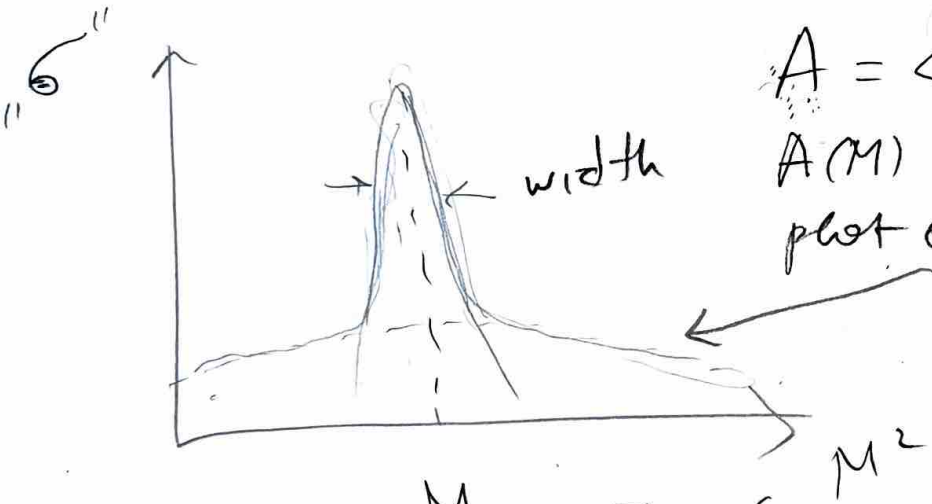
$$M^2 = \left(\sum_{i=1}^{n} P_i\right)^2$$

← Amplitude

$|A|^2$ - probability

$$P_x = P_1 + P_2 + \ldots + P_n$$

$$P_x^2 = (P_1 + P_2 + \ldots + P_n)^2 = M^2$$



$A = \langle P_1, P_2, P_3 \ldots P_n | P_x \rangle$

$A(M)$ then we plot $\sigma \sim |A(M)|^2$

$M_\rho = 770\,(MeV)$

A is an <u>analytical</u> function of variables such as M^2 so, $Re(M) \to$ mass, $Im(M)$ - "width

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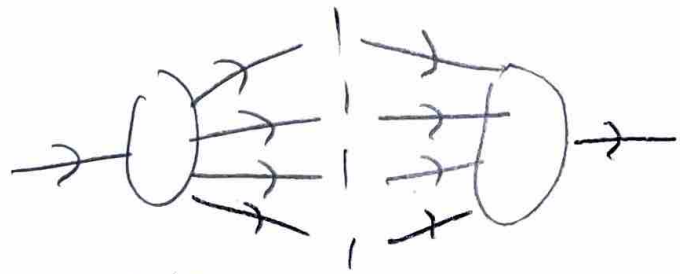
$|A|^2 = ?$



$$\langle P_1 P_2 \ldots P_n | P_x \rangle \langle P_x | P_1 \ldots P_n \rangle$$

$$x = \frac{Q^2}{2\underline{P}\cdot q}$$

$$\underline{P} = (M, \vec{0})$$

$\underline{P}^2 = M^2$, M is the proton mass

$$q = (\nu, \vec{q}), \quad \nu = E_e - E_{e'}$$

let me also define $y = \dfrac{\underline{P}\cdot q}{\underline{P}\cdot e}$ – inelasticity

Try to show that $x \in [0,1]$

$y \in [0,1]$